PROJECT 1 • DSC 201

Edmunds Car Recommender System

Turning 226,931 consumer reviews into relevant, personalized vehicle recommendations

CONTENT-BASED | COLLABORATIVE SVD | HYBRID



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Structured Edmunds Car Reviews Dataset

What problem are we solving?

A single dataset must support content, collaborative, and hybrid recommendation.

THE QUESTION

Which vehicle should this user explore next?

Use what reviewers wrote, how they rated vehicles, and the selected vehicle context.

Grading alignment

- Clear dataset, features, and goal
- Substantial cleaning and preprocessing
- Three required recommender functions
- Efficiency, evaluation, and limitations

01

Content relevance

Find vehicles described with similar review language.

02

Personal preference

Learn user–vehicle patterns from repeated ratings.

03

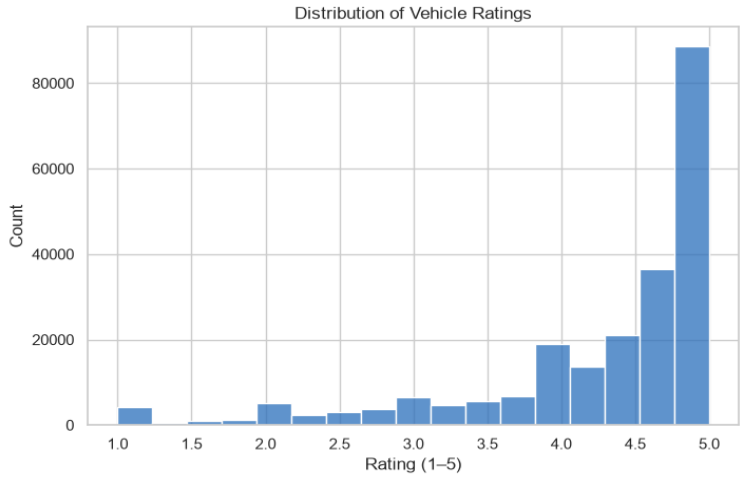
Practical ranking

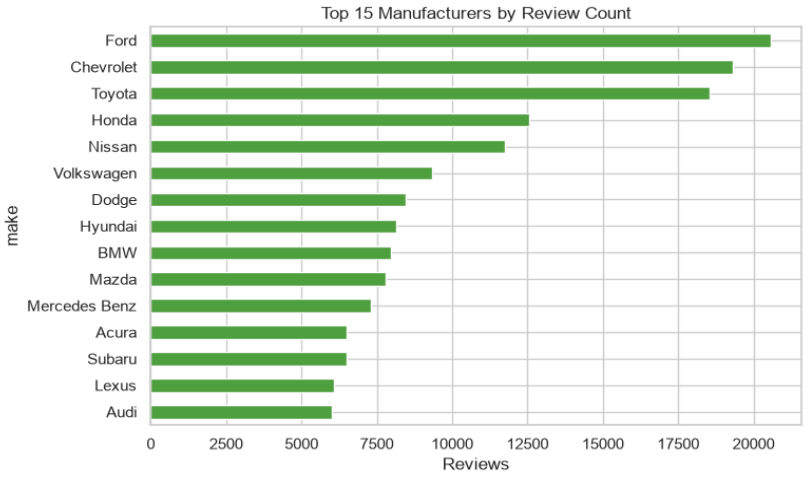
Blend both signals and return useful metadata.

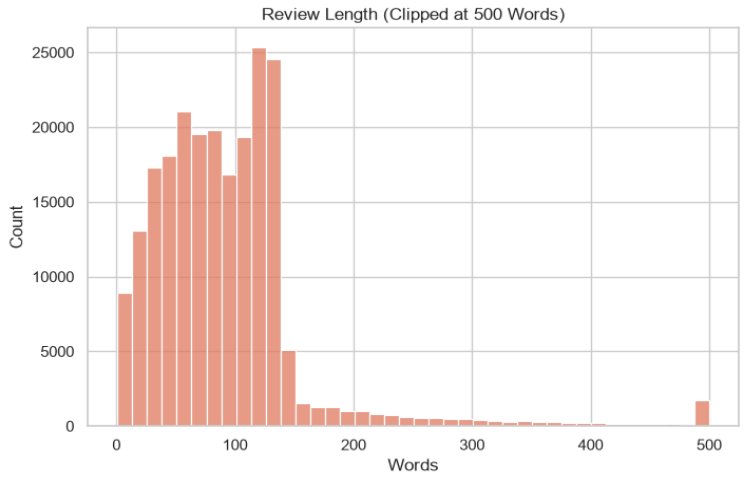
Source: Kaggle • Structured Edmunds Car Reviews Dataset

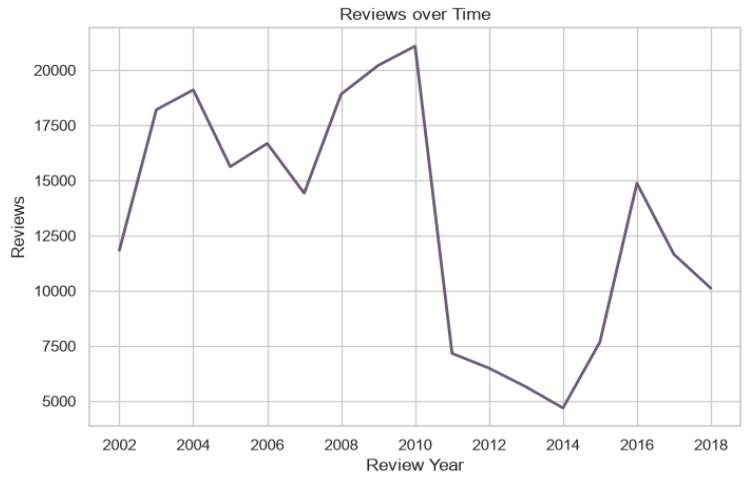
226K reviews across the vehicle market

Text supports content filtering; author–vehicle ratings support collaborative filtering.









226,931
RAW REVIEWS

50
MANUFACTURERS

29,128
VEHICLE TITLES

1–5
RATING RANGE

7
RAW COLUMNS

Author • vehicle title • review title • review text • rating • date • manufacturer

Cleaning was a modeling decision—not a formality

The workflow protects privacy, removes scrape artifacts, and creates stable user–item interactions.

1 Combine

50 manufacturer CSV files

2 Validate

Ratings constrained to 1–5

3 Normalize

Text, dates, makes, model years

4 De-duplicate

Repeated scraped reviews removed

5 Anonymize

Authors mapped to stable user IDs

224,444

CLEAN REVIEWS

2,487 rows removed

155,074

ANONYMOUS USERS

29,039

CLEAN VEHICLES

15,797

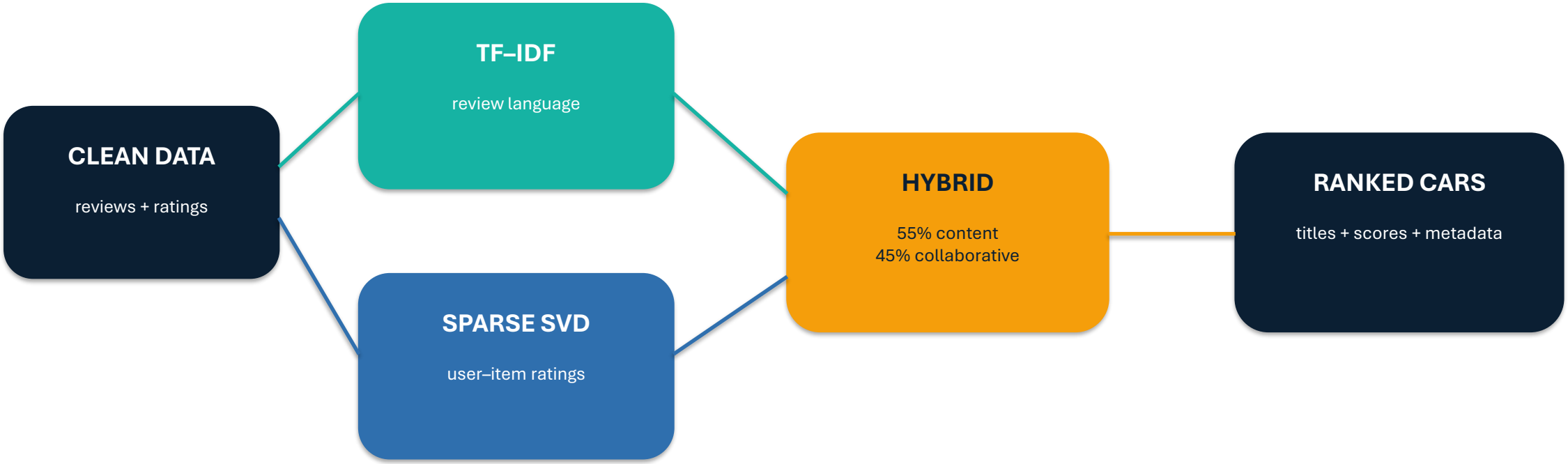
USERS WITH 2+ REVIEWS

Why it matters

Content text becomes comparable; duplicate ratings no longer overweight an item; raw reviewer names never appear in recommendations.

One cleaned dataset → three recommendation paths

Each model answers a different user need; the hybrid re-ranks content candidates with personalization.



`content_based_recommender(title, n)`

`collaborative_recommender(user_id, n)`

`hybrid_recommender(title, user_id, k)`

Required interfaces completed and demonstrated in the executed notebook

Content-based recommendations are coherent and explainable

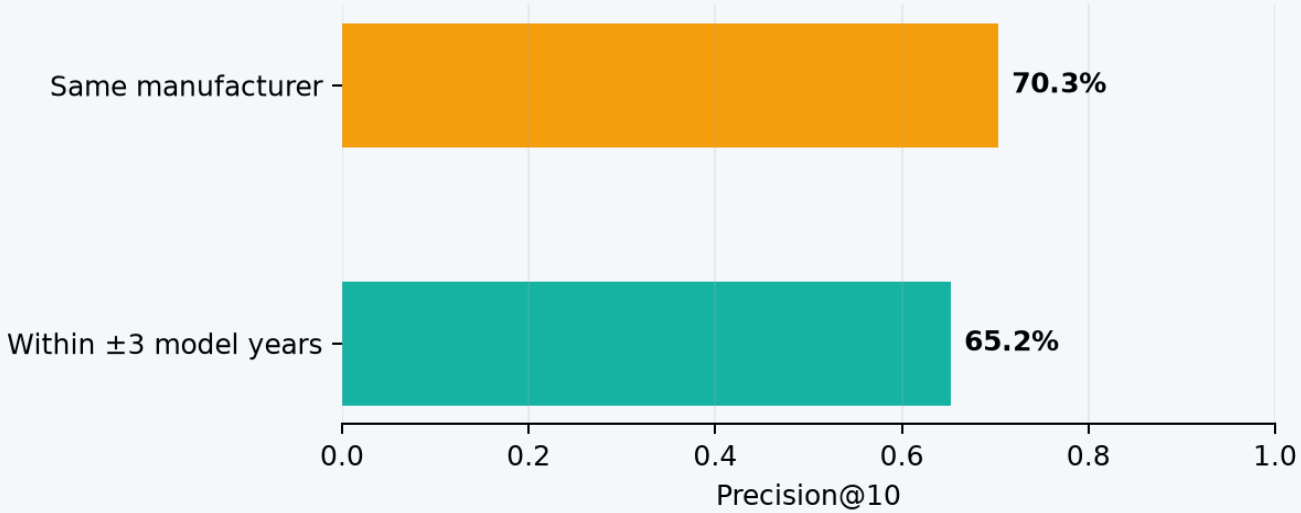
TF–IDF unigrams/bigrams + cosine similarity over aggregated review language.

QUERY VEHICLE

2002 Acura RSX Type-S

Top recommendations

1. 1999 Acura Integra GS-R
2. 2004 Acura RSX Type-S
3. 1998 Acura Integra GS-R
4. 2002 Acura RSX Hatchback
5. 2006 Acura RSX Hatchback



18,287

CONTENT-ELIGIBLE VEHICLES

12,000

TF–IDF FEATURES

Interpretation

The nearest neighbors stay within the same product family and nearby model years—strong evidence that review text captures meaningful vehicle similarity.

Collaborative SVD adds personalization—but sparsity dominates

User-centered sparse SVD is evaluated honestly against simple rating baselines.

26,662
K-CORE INTERACTIONS

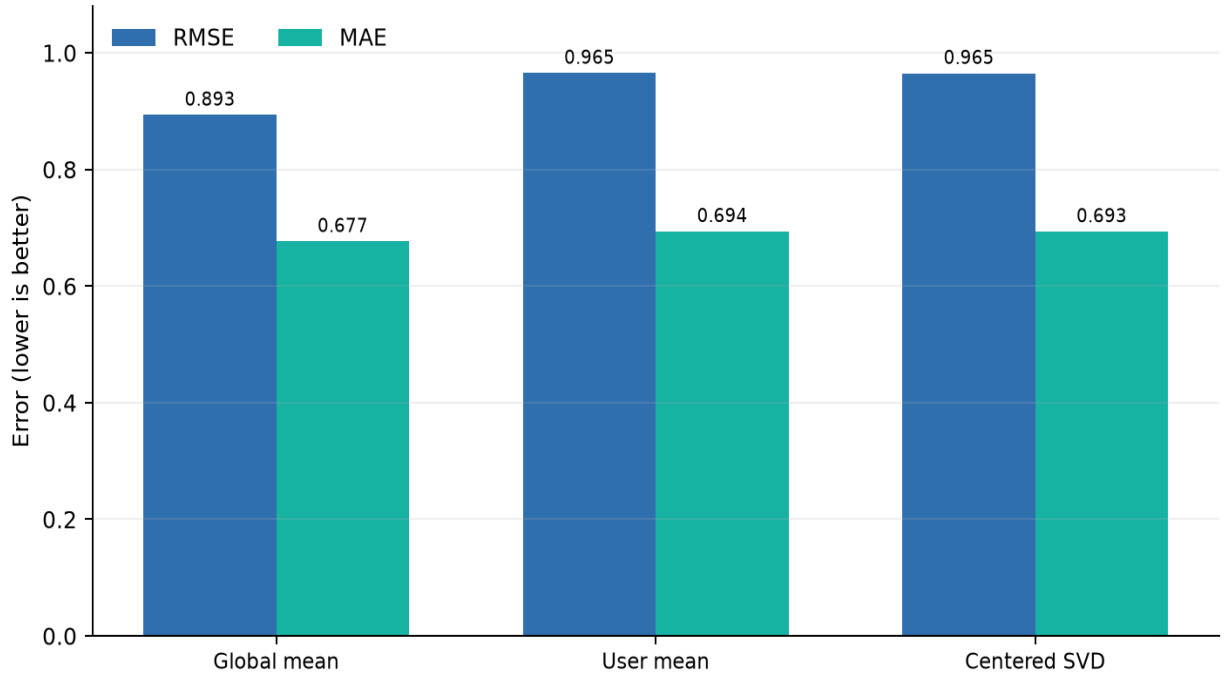
2,378
SUPPORTED USERS

3,384
SUPPORTED VEHICLES

0.302%
MATRIX DENSITY

50
LATENT FEATURES

0.384
EXPLAINED VARIANCE



Honest result

SVD RMSE = 0.965; global-mean RMSE = 0.893. The global average wins because most users rated only one vehicle and ratings are concentrated near 4–5 stars.

Value retained: personalized ranking, not superior rating prediction.

Hybrid ranking balances relevance and preference

Retrieve content neighbors first, then re-rank with the user's SVD score.

HYBRID SCORE

0.55 × content + 0.45 × preference

WHY THIS DESIGN?

Content keeps results relevant to the selected vehicle.

SVD personalizes ordering within that candidate pool.

Previously rated vehicles are excluded.

k controls how many latent features contribute.

≈96 ms

EXAMPLE HYBRID QUERY

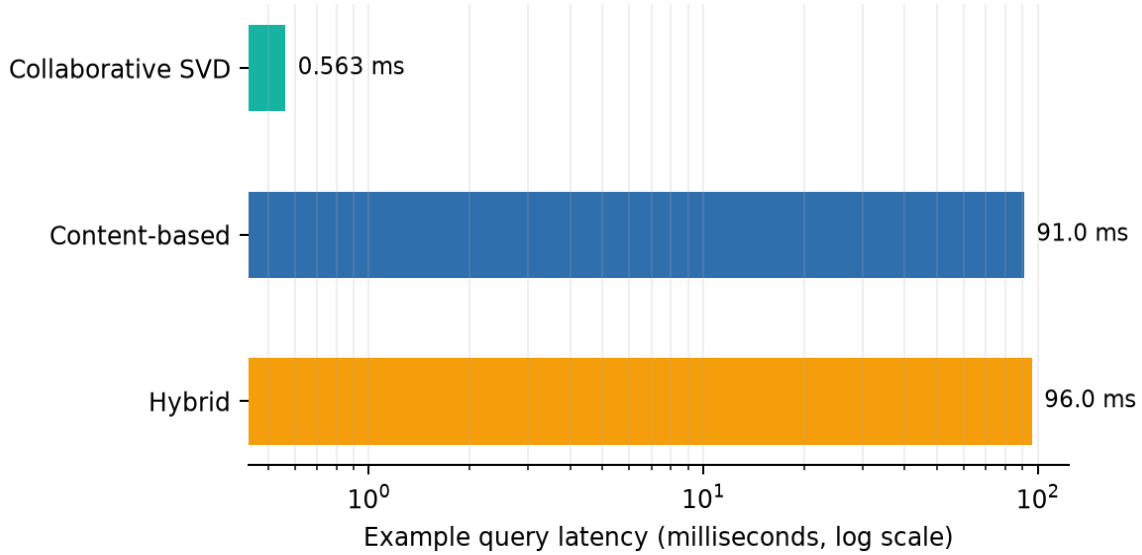
Example ranked output

Vehicle	Year	Pred.	Hybrid
2006 Chevrolet TrailBlazer • SUV • LS 4dr • SUV • (4.2L 6cyl 4A)	2006	3.92	0.889
2006 Chevrolet TrailBlazer • SUV • LT 4dr • SUV • 4WD (4.2L 6cy...	2006	3.92	0.879
2004 Chevrolet TrailBlazer • SUV • LS 4WD 4dr • SUV • (4.2L 6cy...	2004	3.92	0.789
2004 Chevrolet TrailBlazer • SUV • LT 4WD 4dr • SUV • (4.2L 6cy...	2004	3.92	0.752
2005 Chevrolet TrailBlazer • SUV • LT 4WD 4dr • SUV • (4.2L 6cy...	2005	3.92	0.702
2007 Chevrolet TrailBlazer • SUV • LT 4dr • SUV • 4WD (4.2L 6cy...	2007	3.92	0.683

The top results remain closely related while SVD subtly changes their ordering.

Fast queries, different coverage trade-offs

Sparse representations keep all three models usable on a standard Colab runtime.



18,287
CONTENT CATALOG

3,384
COLLABORATIVE CATALOG

9.96%
TOP-10 COVERAGE / 50 USERS

MODEL	INPUT	STRENGTH	LIMITATION
Content	Any supported vehicle	Explainable / cold-start item	No personal signal
SVD	Repeat-user ratings	Very fast personalization	Sparse users and items
Hybrid	Vehicle + user	Best practical balance	Two models + weight choice

What the models can—and cannot—claim

Recommendation quality depends on data coverage, identity quality, and the evaluation target.

CURRENT LIMITATIONS

- Median user has only one review
- Author names are imperfect account identifiers
- Ratings are strongly concentrated near 4–5
- K-core filtering improves reliability but removes rare vehicles
- RMSE does not measure diversity or user satisfaction

NEXT ITERATION

- Add explicit onboarding preferences for new users
- Tune hybrid weights using validation or user studies
- Track precision@k, recall@k, diversity, and novelty
- Monitor popularity and manufacturer exposure
- Retrain as new vehicle reviews arrive

Key principle **A useful recommender is not only accurate—it is relevant, diverse, fast, explainable, and honest about uncertainty.**

CONCLUSION

The hybrid model is the strongest practical answer

Content provides coherent candidates. SVD adds user preference. The hybrid keeps both signals visible and measurable.

70.3%	18,287	≈96 ms
same-make precision@10	content catalog vehicles	hybrid example query

Deliverables completed

✓ content_based_recommender(title, n) ✓ collaborative_recommender(user_id, n) ✓ hybrid_recommender(title, user_id, k)

Questions?

GitHub: github.com/tarun-vignes/DSC201-Project-I